

*Novel Methods to Evaluate Trauma
Treatments in Research and Care:
Bayesian and IC Evaluation of Informative
Hypotheses*

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BETTR: Bayesian Evaluation of Treatments for Traumatized Refugees

- Develop interventions for traumatized groups underrepresented in research.
- Evaluate the effects of these interventions in single-case experimental designs.
- Refine and apply Bayesian and Information Theory methods
- Create accessible statistical tools for clinicians and researchers to evaluate their interventions – both in clinical and research settings.

Agenda

1. Background (15 min)

- Individual Participant Data
- Informative Hypotheses
- Example

2. Practice (30 min)

- Formulating hypotheses
- Analysis in JASP

3. Q&A and Resources (15 min)

Goals

- 1) Learn how to translate clinical questions into hypotheses,
- 2) Learn how to evaluate hypotheses and interpret analytical results

Data in Clinical Settings

- In trauma treatment, patients often present with co-
occurring conditions, chronic symptoms, and a combination of mental health issues and *psychosocial difficulties*.

In these cases, large-scale randomized controlled trials to assess innovative interventions are often not feasible.

Individual Participant Data

We can use IPD to evaluate newly developed interventions in small, specific patient groups.

Valuable insights: capture individual symptom changes, variance, and treatment trajectories.

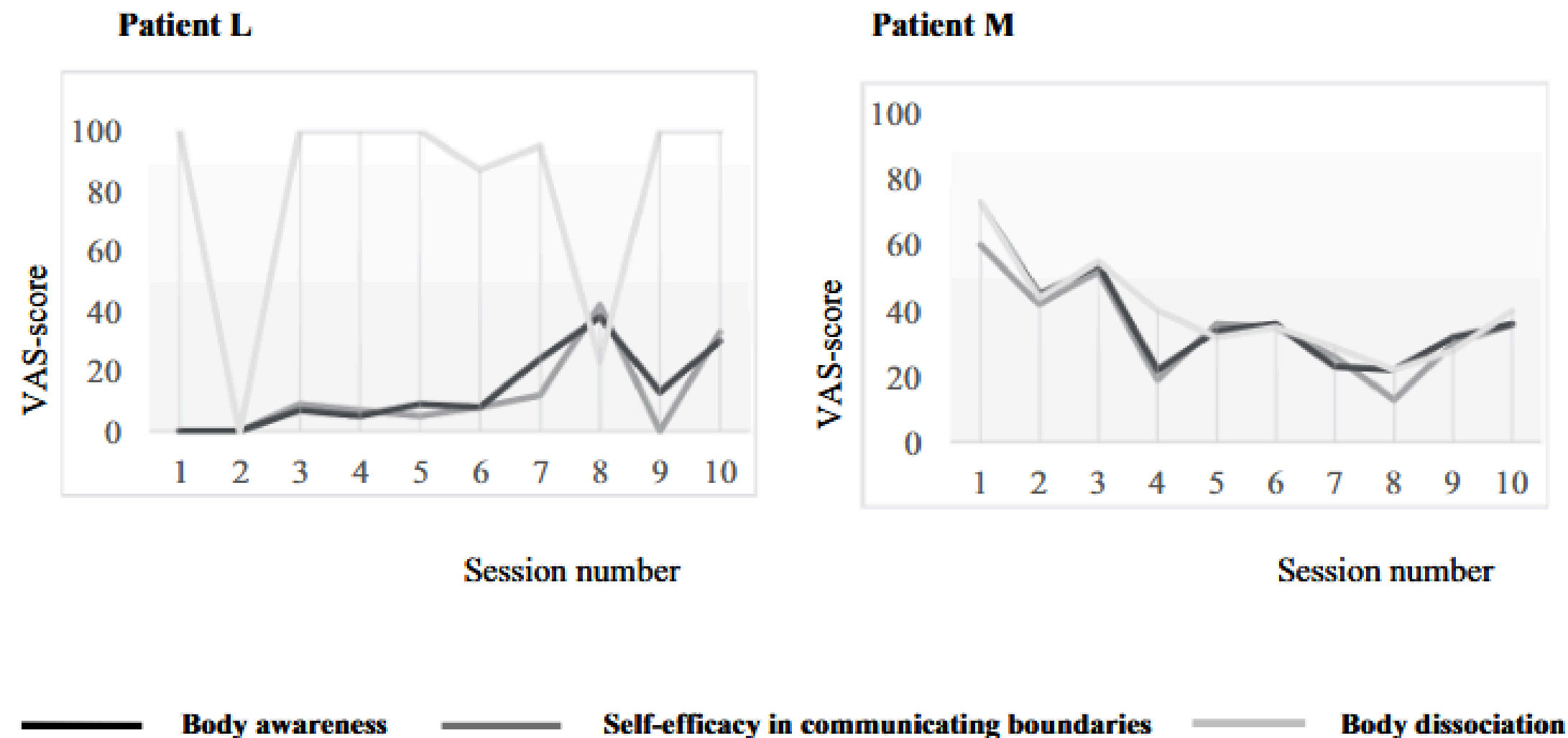


Figure 2. (Continued).

Informative Hypotheses

How to evaluate treatment outcomes?

- Most researchers and clinicians have theory-based (or experience-based) expectations about what works, for whom, and in which direction.
- Using informative hypotheses allows us to directly quantify support in favor of (or against) these expectations

An Example:

Do symptoms of Self-Efficacy, Body Awareness, and Body Dissociation improve during treatment?

Two patterns of interest to evaluate:

- Hypothesis 1: scores on all three outcomes don't change
- Hypothesis 2: scores on all three outcomes improve
- Fail-safe hypothesis: the complement of 1 and 2

Bayes Factor

Posterior Model Probabilities

GORICA weights

An Example:

Do symptoms of Self-Efficacy, Body Awareness, and Body Dissociation improve during treatment?

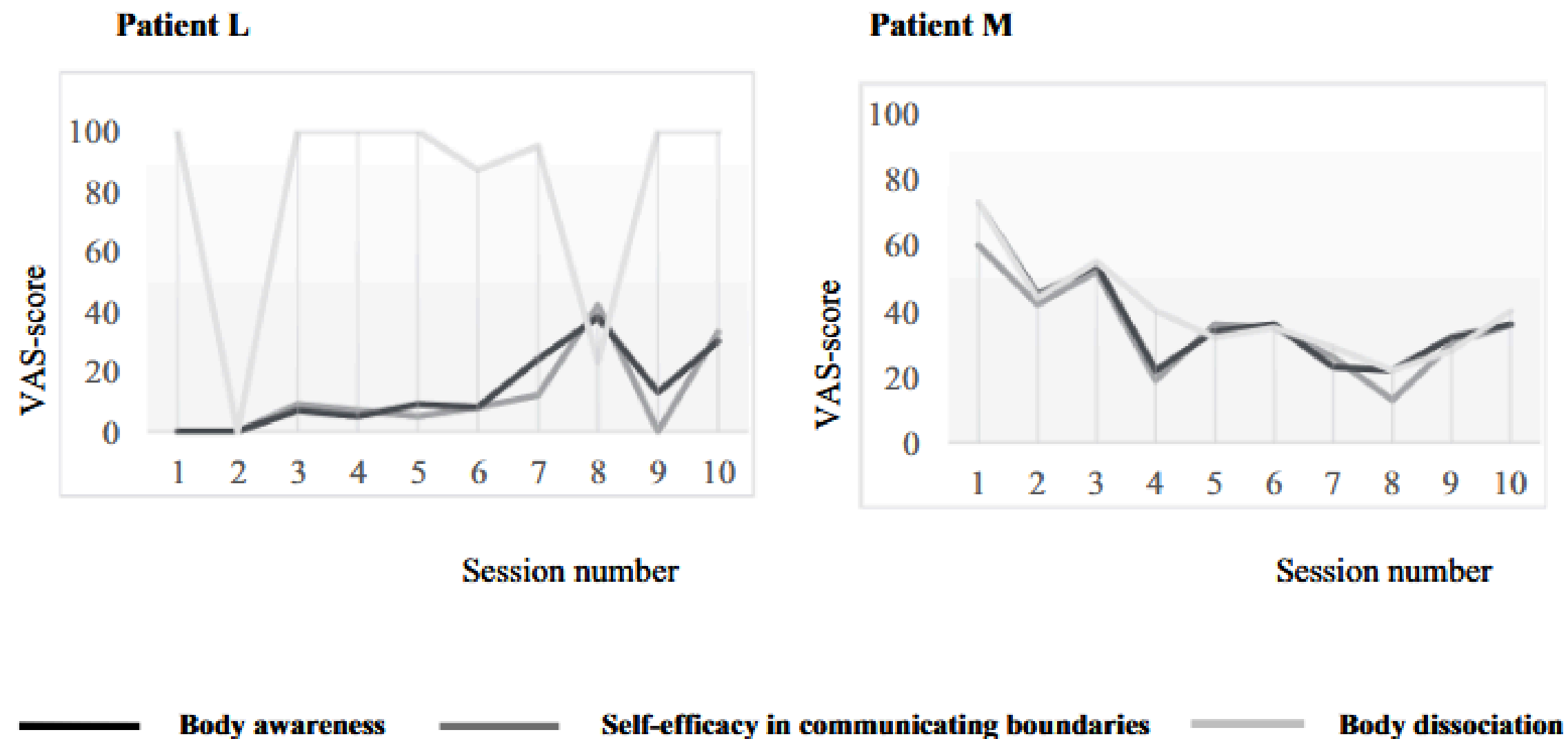


Figure 2. (Continued).

PRACTICE

Dataset

Fictitious case:

- 35-year-old refugee patient
- Diagnosed with PTSD and PGD
- Culturally-Adapted Trauma-Focused CBT
- 24 treatment sessions
- PTSD, PGD, and depressive symptoms assessed weekly

Dataset

Fictitious case:

- 35-year-old refugee patient
- Diagnosed with PTSD and PGD
- Culturally-Adapted Trauma-Focused CBT
- 24 treatment sessions
- PTSD, PGD, and depressive symptoms assessed weekly

Which research questions would *you* be interested in?

Research Question & Hypotheses

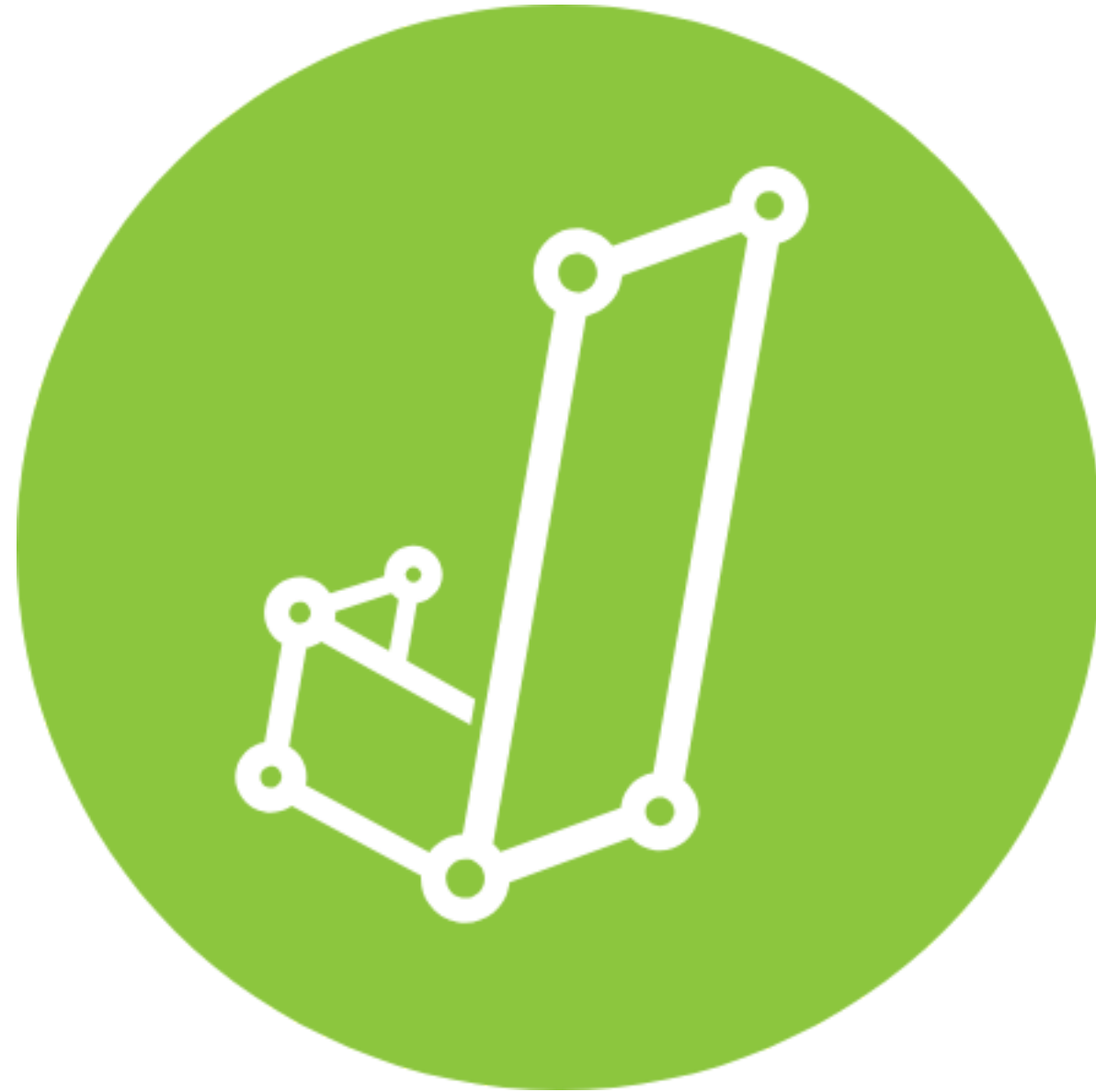
RQ: Do symptoms of PTSD, PGD, and Depression improve during treatment?

Research Question & Hypotheses

RQ: Do symptoms of PTSD, PGD, and Depression improve during treatment?

We define two patterns of interest:

- Hypothesis 1: scores on all three outcomes don't change
- Hypothesis 2: scores on all three outcomes improve
- Fail-safe hypothesis: the complement of 1 and 2



JASP

New Data

Descriptives

T-Tests

ANOVA

Mixed Models

Regression

Frequencies

Factor

Bain

New

Open

Save

Save As

Export Results

Export Data

Sync Data

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Browse

two phase improvement
/Users/ /SCED JASP/

one phase improvement
/Users/ /SCED JASP/

Data Sets
/Applications/JASP.app/Contents/Resources/

Downloads
/Users/

Desktop
/Users/

JASP 0.96 (Apple Silicon)

P

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munity

Ask a question
Visit the website

patient1_data*

Edit Data

Descriptives

T-Tests

ANOVA

Mixed Models

Regression

Frequencies

	t_measurement	PTSD	PGD	DEP	
1	1	20	16	26	
2	2	15	13	14	
3	3	15	21	16	
4	4	8	19	14	
5	5	18	13	19	
6	6	15	11	14	
7	7	16	13	17	
8	8	16	19	12	
9	9	15	17	17	
10	10	11	9	16	
11	11	18	14	18	
12	12	2	10	14	
13	13	11	13	20	
14	14	17	13	13	
15	15	11	10	20	
16	16	16	10	16	
17	17	14	10	13	
18	18	12	10	10	
19	19	3	5	21	
20	20	10	-2	11	
21	21	16	11	17	
22	22	13	4	18	
23	23	6	2	17	
24	24	4	2	13	

patient1_data*

Edit Data

Descriptives

T-Tests

ANOVA

Mixed Models

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Frequencies

Factor

Bain

t_measurement

PTSD

PGD

DEP

1	1	20	16	26
2	2	15	13	14
3	3	15	21	16
4	4	8	19	14
5	5	18	13	19
6	6	15	11	14
7	7	16	13	17
8	8	16	19	12
9	9	15	17	17
10	10	11	9	16
11	11	18	14	18
12	12	2	10	14
13	13	11	13	20
14	14	17	13	13
15	15	11	10	20
16	16	16	10	16
17	17	14	10	13
18	18	12	10	10
19	19	3	5	21
20	20	10	-2	11
21	21	16	11	17
22	22	13	4	18
23	23	6	2	17
24	24	4	2	13

Select channel:

☒ Official

☐ Community

☒ Show Betas

Search for a module:

Acceptance Sampling

Official

Sampling for acceptance

Installed 0.95.5-release.12 by jasp-stats

Latest stable 0.95.5-release.12 on 28/02/2026 with 142 downloads

Latest beta 0.96.1-beta.1 on 11/04/2026 with 52 downloads

Update Beta

ANOVA

Official

Evaluate the difference between multiple means

Installed 0.95.5-release.12 by jasp-stats

Latest stable 0.95.5-release.12 on 28/02/2026 with 135 downloads

Latest beta 0.96.3-beta.0 on 14/04/2026 with 14 downloads

Update Beta

Update All (36)

Install Local Module

Show modules menu

Install Developer Module

☐ Acceptance Sampling

☐ Addit

☒ Bain

☐ Bayes Factor Functions

☐ BFpack

☐ BSTS

☐ Circular Statistics

☐ Cochrane Meta-Analyses

☐ Distributions

☐ Equivalence T-Tests

☐ ESCI

☐ JAGS

☐ Learn Bayes

☐ Learn Stats

patient1_data*

Edit Data

Descriptives

T-Tests

ANOVA

Mixed Models

Regression

Frequencies

Factor

Bain

t_measurement

PTSD

PGD

DEP

+

1	1	20	16	26
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16	16	16	10	16
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19	19	3	5	21
20	20	10	-2	11
21	21	16	11	17
22	22	13	4	18
23	23	6	2	17
24	24	4	2	13

T-Tests

Welch's T-Test

Paired Samples T-Test

One Sample T-Test

ANOVA

ANOVA

ANCOVA

Regression

Linear Regression

Structural Equation Modeling

▼ Bain Structural Equation Modeling



|

⌘ + Enter to apply

Model Options

Factor scaling

Grouping variable

Tables

☒ Coefficients

Credible interval %

Plots

☐ Path diagram

☐ Show parameter estimates

☐ Show legend

Additional Options

Seed

Fraction

☐ Standardize coefficients

Tables

☐ Bayes factor matrix

Plots

☐ Posterior probabilities

► Model Constraints

▼ Bain Structural Equation Modeling



```
PTSD~t_measurement  
PGD~t_measurement  
DEP~t_measurement|
```

⌘ + Enter to apply

▼ Bain Structural Equation Modeling



```
PTSD~t_measurement  
PGD~t_measurement  
DEP~t_measurement
```

⌘ + Enter to apply

Model Options

Factor scaling Factor variance ▼

Grouping variable <no choice> ▼

Tables

☒ Coefficients

Credible interval 95 %

Plots

☐ Path diagram

☐ Show parameter estimates

☐ Show legend

Additional Options

Seed 529778

Fraction 1

☐ Standardize coefficients

Tables

☐ Bayes factor matrix

Plots

☐ Posterior probabilities

▶ Model Constraints

▼ Model Constraints

Hypotheses can be formulated using the parameter names that appear if the 'Coefficients' box is ticked. Do NOT use spaces in the parameter names. Place each hypothesis on a new line. For example:

$A_{\sim x1} = A_{\sim x2} = A_{\sim x3} = A_{\sim x4}$

$A_{\sim x1} > A_{\sim x2} > A_{\sim x3} = A_{\sim x4}$

$A_{\sim x1} > 0 \ \& \ A_{\sim x2} > 0 \ \& \ A_{\sim x3} > 0 \ \& \ A_{\sim x4} > 0$

Read the help file for further instructions about formulating hypotheses and what can and cannot be done with Bain Structural Equation Modeling.

|

⌘ + Enter to apply

▼ Model Constraints

Hypotheses can be formulated using the parameter names that appear if the 'Coefficients' box is ticked. Do NOT use spaces in the parameter names. Place each hypothesis on a new line. For example:

$A \sim x_1 = A \sim x_2 = A \sim x_3 = A \sim x_4$

$A \sim x_1 > A \sim x_2 > A \sim x_3 = A \sim x_4$

$A \sim x_1 > 0 \ \& \ A \sim x_2 > 0 \ \& \ A \sim x_3 > 0 \ \& \ A \sim x_4 > 0$

Read the help file for further instructions about formulating hypotheses and what can and cannot be done with Bain Structural Equation Modeling.

```
PTSD~t_measurement < 0 & PGD~t_measurement < 0 & DEP~t_measurement < 0  
PTSD~t_measurement = 0 & PGD~t_measurement = 0 & DEP~t_measurement = 0
```

⌘ + Enter to apply

Bain Structural Equation Modeling ▼

Hypothesis Legend ▼

Hypothesis	
H1	PTSD~t_measurement < 0 & PGD~t_measurement < 0 & DEP~t_measurement < 0
H2	PTSD~t_measurement = 0 & PGD~t_measurement = 0 & DEP~t_measurement = 0

Bain Structural Equation Model

	BF.u	BF.c	PMP a	PMP b	PMP c
H1	5,492	30,01	1,000	0,846	0,968
H2	$6,194 \times 10^{-9}$	$6,194 \times 10^{-9}$	$1,128 \times 10^{-9}$	$9,540 \times 10^{-10}$	$1,091 \times 10^{-9}$
Hu				0,154	
Hc	0,183				0,032

Note. BF.u denotes the Bayes factor of the hypothesis at hand versus the unconstrained hypothesis Hu. BF.c denotes the Bayes factor of the hypothesis at hand versus its complement. PMPa contains the posterior model probabilities of the hypotheses specified. PMPb adds Hu, the unconstrained hypothesis. PMPc adds Hc, the complement of the union of the hypotheses specified. All PMPs are based on equal prior model probabilities.

Coefficients for Parameters

	N	Mean	SE	95% Credible Interval	
				Lower	Upper
PTSD~t_measurement	24	-0,341	0,125	-0,585	-0,097
PGD~t_measurement	24	-0,640	0,099	-0,835	-0,446
DEP~t_measurement	24	-0,104	0,102	-0,304	0,095

Note. The displayed coefficients are unstandardized.

The hypotheses we specified →

The BF and PMP of the hypotheses →

The coefficient estimates →

Hypothesis Legend ▼

Hypothesis	
H1	PTSD~t_measurement < 0 & PGD~t_measurement < 0 & DEP~t_measurement < 0
H2	PTSD~t_measurement = 0 & PGD~t_measurement = 0 & DEP~t_measurement = 0

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The image shows a screenshot of a statistical software interface. At the top, there is a toolbar with icons for 'Edit Data', 'Descriptives', 'T-Tests', 'ANOVA', and 'Mixed'. The 'Descriptives' icon, which shows a bar chart, is highlighted with a yellow rectangle. A dropdown menu is open from the 'Descriptives' icon, listing the following options: 'Descriptive Statistics', 'Raincloud Plots', 'Time Series Descriptives', 'Flexplot' (which is highlighted), and 'Plot Builder (beta)'. Below the toolbar, the main panel is titled 'Bain Structural' with a dropdown arrow. Under this title, there are three lines of text: $PTSD \sim t_measurement$, $PGD \sim t_measurement$, and $DEP \sim t_measurement$. On the left side of the main panel, there are three vertical dots indicating a sidebar or menu.

Edit Data

Descriptives

T-Tests

ANOVA

Mixed

▼ **Bain Structural**

$PTSD \sim t_measurement$

$PGD \sim t_measurement$

$DEP \sim t_measurement$

Descriptive Statistics

Raincloud Plots

Time Series Descriptives

Flexplot

Plot Builder (beta)

Flexplot - PTSD

PGD
DEP



Dependent Variable

PTSD

Independent Variable(s)

t_measurement

Paneled Variable(s)

Options

Point controls

Point transparency Jitter in X Jitter in Y



0,4



0,1



0

Visual Statistics



Fitted line (scatterplots)

Regression ▼

Intervals (categorical predictors) Quantiles

Other Plot Controls

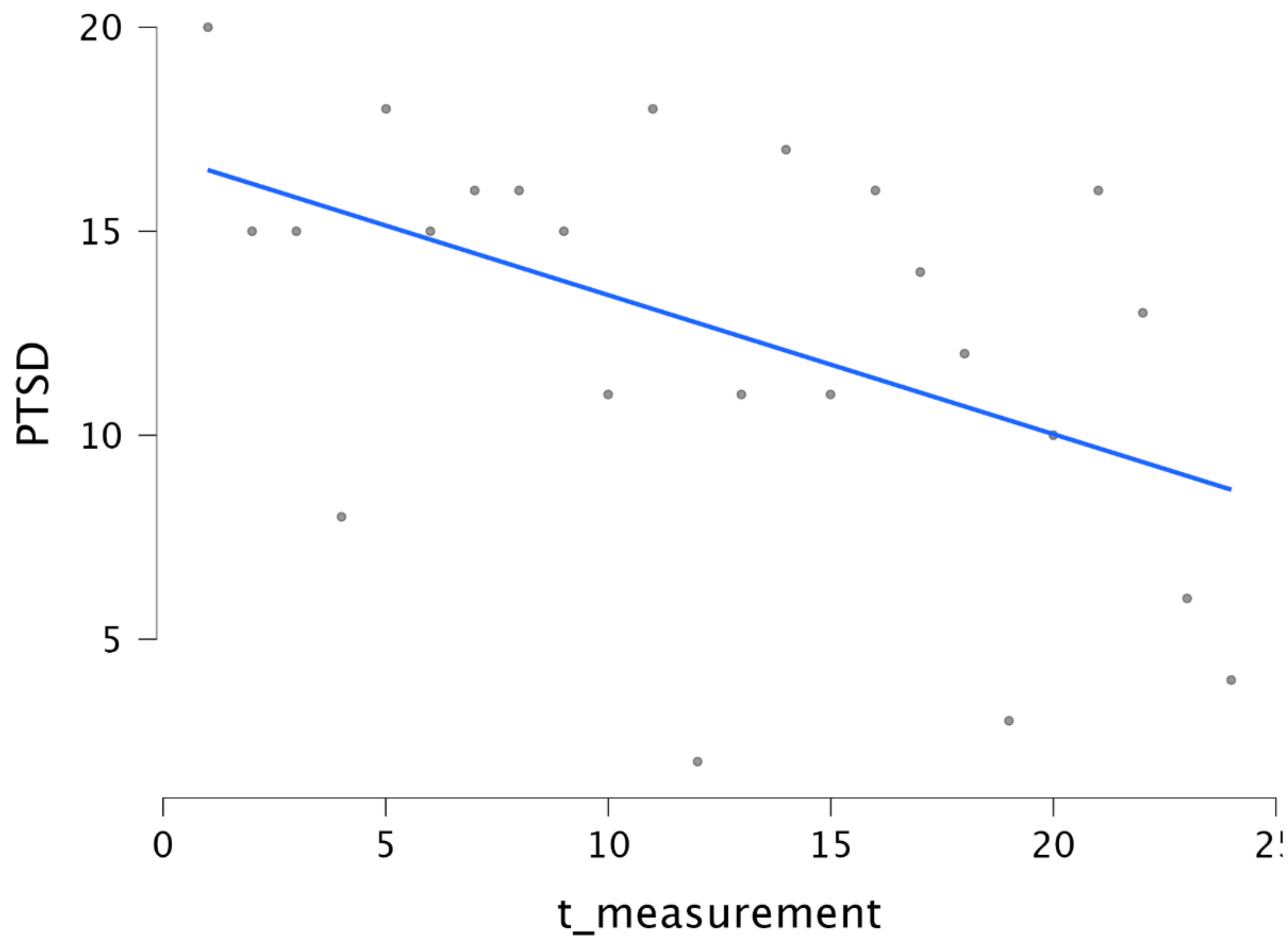
GGplot theme JASP ▼

Color Palette GGplot Default ▼

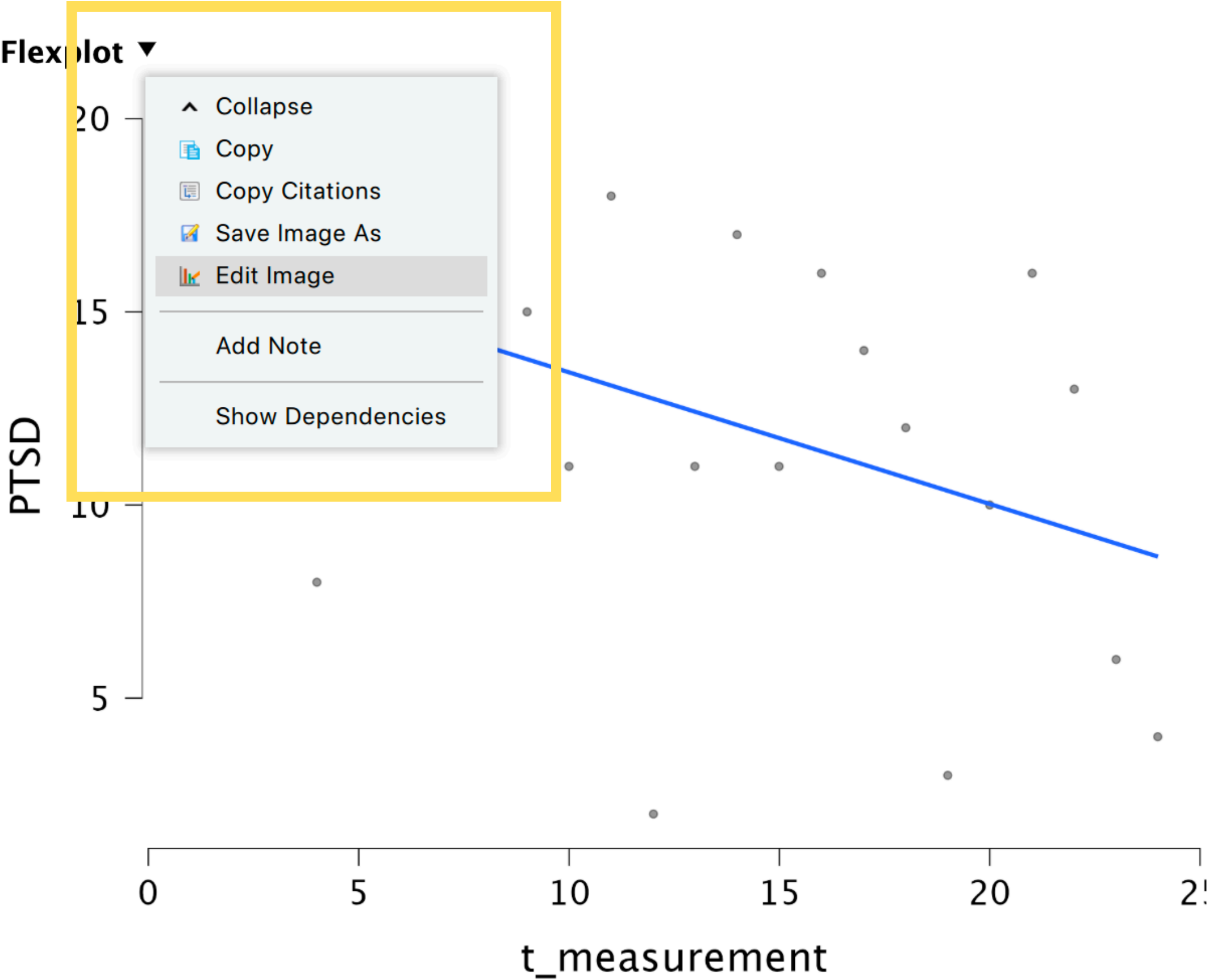
☐ Convert to grayscale

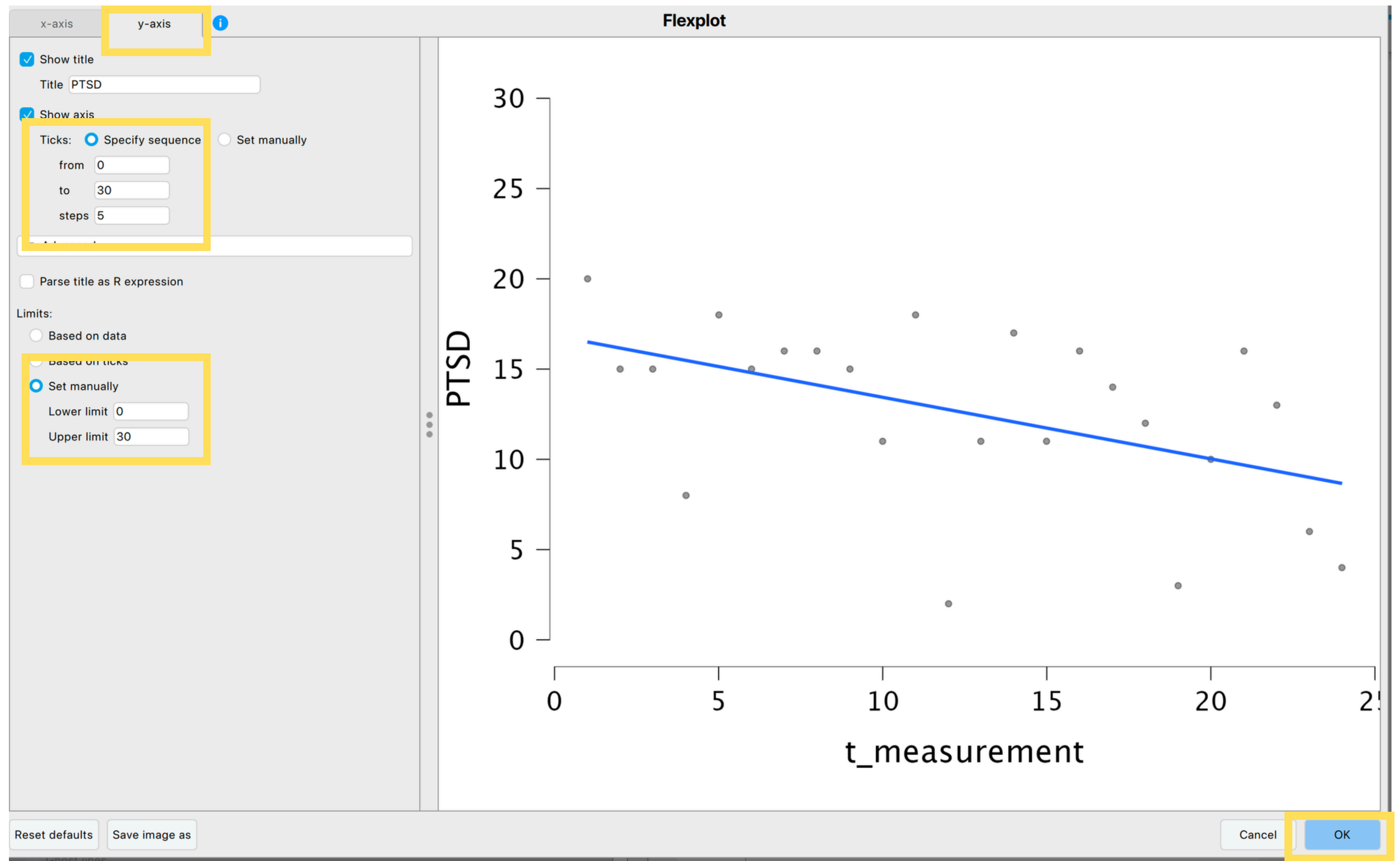
Flexplot – PTSD ▼

Flexplot



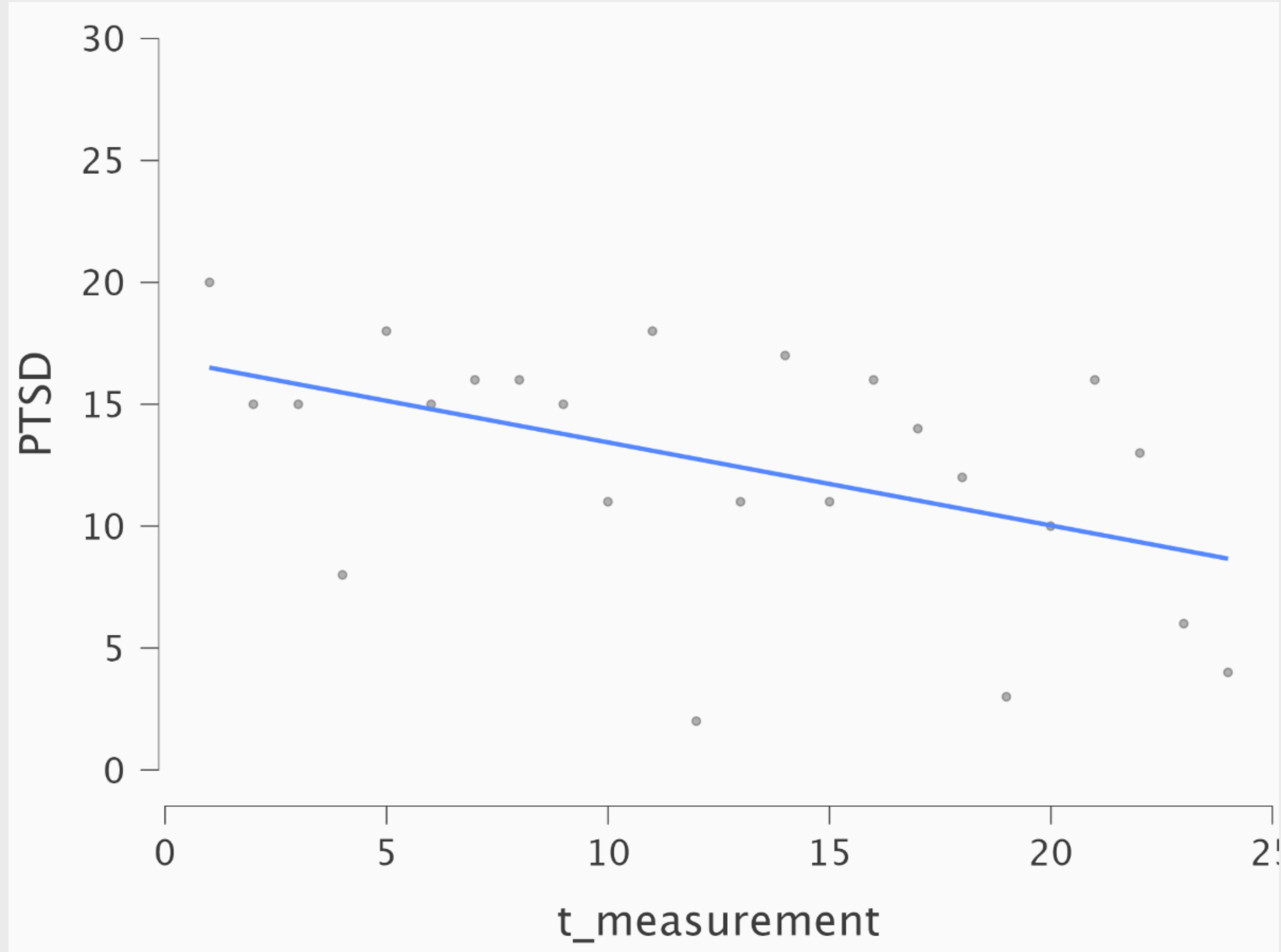
Flexplot – PTSD





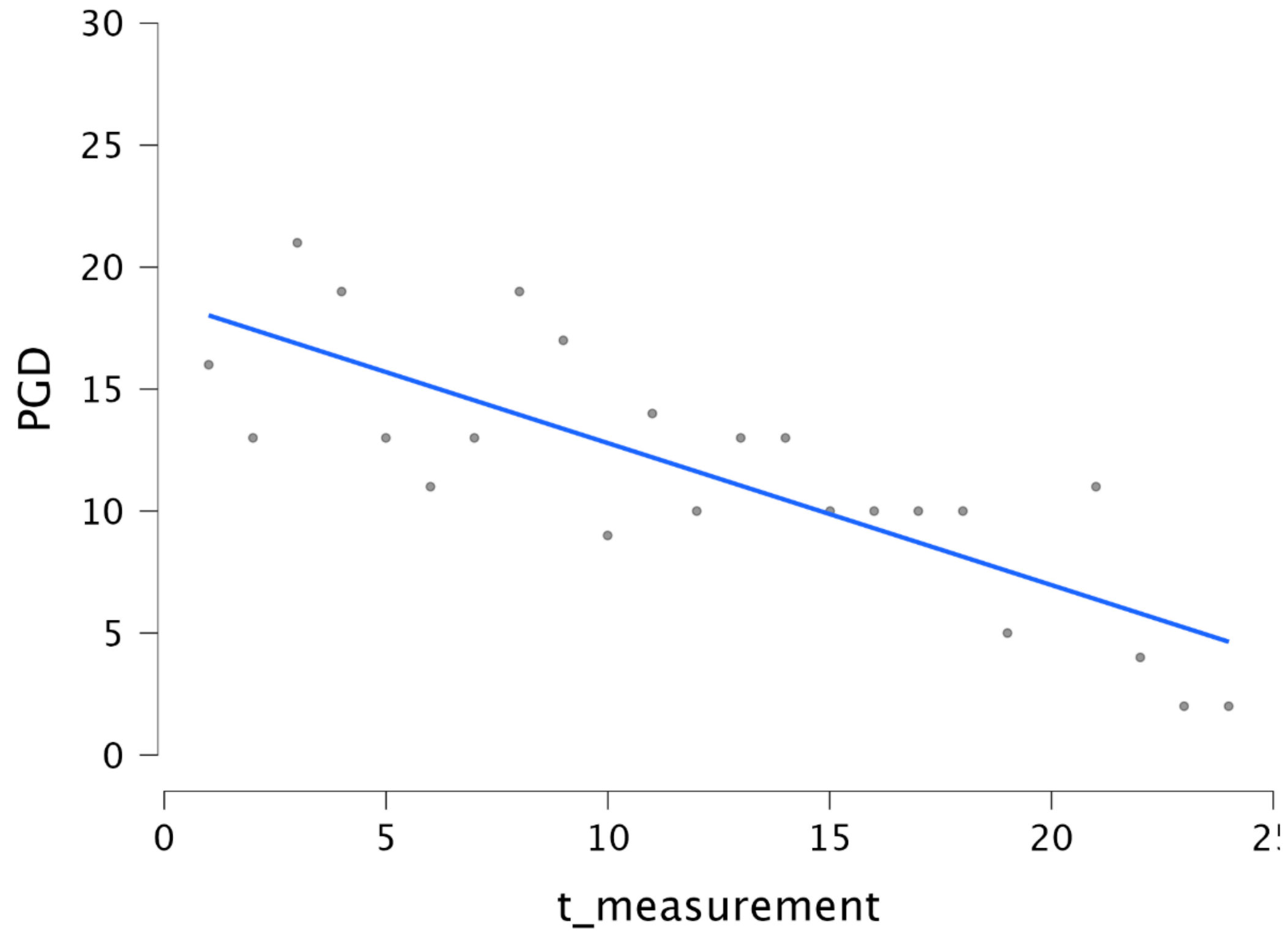
Flexplot - PTSD

Flexplot



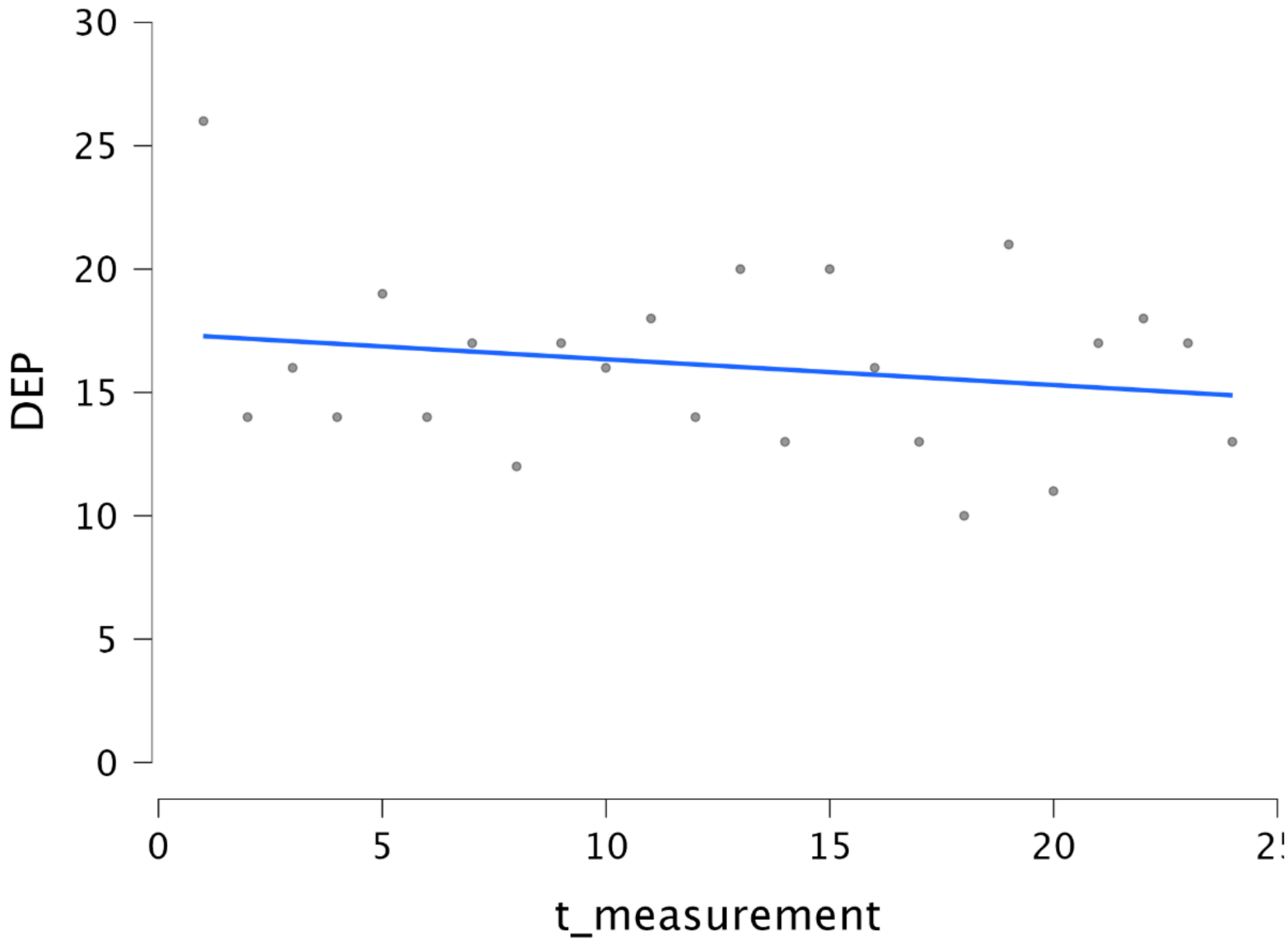
Flexplot – PGD

Flexplot



Flexplot - DEP ▼

Flexplot ▼



H1: PTSD < 0 ; PGD < 0 ; DEP < 0
H2 = Heq: PTSD = 0 ; PGD = 0 ; DEP = 0

restriktor (0.6-30): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights	gorica.weights_without_heq
1	Heq	-16.343	0.000	32.686	0.000	0.782	0.000	
2	H1	3.841	1.608	-4.466	0.618	0.157	0.805	0.805
3	complement	3.358	2.543	-1.632	0.382	0.062	0.195	0.195

Conclusion:

- The order-restricted hypothesis H1 is the best in the set, as it has the highest GORIC(A) weight. Since it has a higher GORIC(A) weight than the equality-restricted hypothesis (Heq), we can now inspect the relative support for the order-restricted hypothesis against its complement:

* 'H1' is 4.125 times more supported than 'complement'.

Results

Table 1

Per Hypothesis, Posterior Model Probability (PMP), GORICA weight, and the Bayes Factor (BFc)

Hypothesis	PMP	GORICA Weight	<u>BFc</u>
Informative Hypothesis (H1)	0.97	0.81	30.01
Equality Hypothesis (H2)	0.00	0.00	0.00
Complement (<u>Hc</u>)	0.03	0.19	

Note. Hypotheses: H1: $\beta_{1,PTSD} < \beta_{1,PGD} < \beta_{1,Dep} = 0$; H2: $\beta_{1,PTSD} = 0$ & $\beta_{1,PGD} = 0$ & $\beta_{1,Dep} = 0$; Hc: not H1 and not H2.

Reporting Example: PMP and GORICA weights

The results of our analyses indicate PMPs of 0.00 for the equality hypothesis H_2 , 0.03 for the complement H_c , and 0.97 for H_1 . Based on the observed data and the set of hypotheses, there is very strong evidence in favor of H_1 . Similarly, the GORICA weight for H_1 was 0.81, indicating it was the most supported of the three hypotheses, given the observed data. The equality hypothesis H_2 received virtually no support with a GORICA weight of 0.00. The complement H_c shows weak support with a GORICA weight of 0.19. The ratio of the GORICA weights of H_1 and H_c indicates that H_1 is 4.13 times more supported than the complement. While Bayesian evaluation suggests very strong evidence in favor of H_1 , the GORICA weights suggest strong evidence in favor of H_1 but also weak evidence in favor of H_c .

Overall, the informative hypothesis H_1 , stating that symptoms decrease over the course of treatment, was supported by the data. For this specific patient we conclude that symptoms improved over the course of treatment.

Thank you for your attention!

Questions?



Open for collaboration

Contact us:

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RESOURCES

- **Website** (will be live soon!): <https://www.uu.nl/en/research/the-BETTR-project>
- **GitHub**: https://github.com/gorizzon/BETTR/tree/main/NtVP_Workshop



**Utrecht
University**

Sharing science,
shaping tomorrow

Contact us:

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